Morning Quiz - Answer Key

PV101 Day 5 - Monday, July 13, 2026

Name: _____ Date: _____

Use notes

Suggested answers are below. Accept equivalent plain-language wording when the student clearly understands the safety concept.

1. In a solar/battery circuit, what does voltage describe?

Electrical pressure or potential difference. It tells us how hard the circuit can push charges.

2. What does current describe?

Flow of electric charge, measured in amps.

3. What is the formula for power in watts?

P = V x A.

4. A 12.8 V battery is supplying 5 A to a load. About how many watts is that?

12.8 x 5 = 64 W.

5. A 600 W inverter load on a 12 V battery can require roughly how much current before losses?

600 / 12 = 50 A before inverter losses, so real current may be higher.

6. In a series connection, what usually adds: voltage or current capacity?

Voltage adds. Current is limited by the path/components.

7. In a parallel connection, what usually stays the same and what increases?

Voltage stays the same. Current capacity can increase.

8. What is a fuse actually protecting?

The wire/conductor from overheating during too much current, not the device’s feelings.

9. Where should the main load/inverter fuse be placed in a small battery system?

Close to the battery positive terminal, before the long wire run to the load/inverter.

10. Why do we measure battery voltage before connecting equipment?

To verify battery state/polarity and catch bad assumptions before energizing the system.

11. What is voltage drop?

Loss of voltage along a wire or connection while current is flowing.

12. Name one sign that a wire, crimp, or connection should be treated as unsafe.

Heat, smell, discoloration, sparks, loose terminal, exposed copper, high voltage drop, intermittent reading, or failed tug test.